Fuel Consumption and Route Comparison (Case Study 18)

Problem Statement

Calculate the fuel required and travel cost for a journey using distance, mileage, and fuel price.

Objectives

Practise arithmetic, variables, and comparisons.

Assumptions

Distance, mileage, and fuel price are positive numbers. Reserve fuel is not included in this simple version.

Algorithm

1. Read distance, mileage, and price per litre. 2. Fuel needed = distance / mileage. 3. Cost = fuel needed × price. 4.
Display both values.

Flowchart (Text Form)

START → Read distance, mileage, price → Validate positive values → Calculate litres and cost → Print results → STOP.

Pseudocode

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Python Source Code

```
distance = float(input("Journey distance (km): "))
mileage = float(input("Vehicle mileage (km/litre): "))
price = float(input("Fuel price per litre: "))
if distance &lt;= 0 or mileage &lt;= 0 or price &lt;= 0:
    print("Enter positive values")
else:
    fuel = distance / mileage
    cost = fuel * price
    print("Fuel required:", round(fuel, 2), "litres")
    print("Journey cost: Rs.", round(cost, 2))
```

Test Case and Expected Output

Input: distance=150 km, mileage=15 km/litre, price=100
Expected: Fuel required: 10.0 litres; Journey cost: Rs. 1000.0

Conclusion

The program estimates fuel use and cost for a journey using simple calculations.